Boosting

Gradient Boosting: CWB Basics 1

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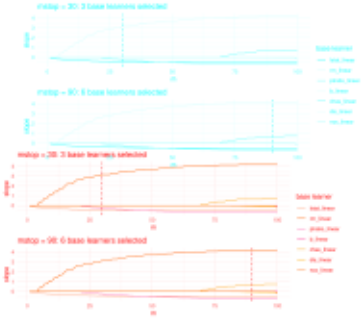

Learning goals

- Concept of CWB
- Which base learners do we use
- Built-in feature selection

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COMPONENTWISE GRADIENT BOOSTING

GB (with trees), has strong predictive performance but is difficult to interpret unless the base learners are stumps.

The aim of CWB is to find a model that exhibits:

- strong predictive performance,
- interpretable components,
- automatic selection of components,
- is sparser than a model fitted with maximum-likelihood estimation.

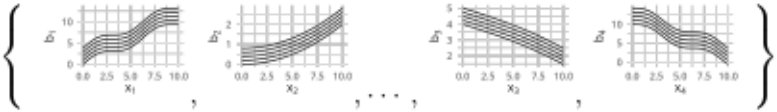
This is achieved by using “nice” base learners which yield familiar statistical models in the end.

Because of this, CWB is also often referred to as **model-based boosting**.

BASE LEARNERS

In GB only one kind of base learner $\mathcal{B}$ is used, e.g., regression trees.

For CWB we generalize this to multiple base learner sets $\{\mathcal{B}_1, ...\mathcal{B}_J\}$ with associated parameter spaces $\{\boldsymbol{\Theta}_1, ...\boldsymbol{\Theta}_J\}$, where $j \in \{1, 2, \ldots, J\}$ indexes the type of base learner.



In each iteration, base learners are fitted to the **pseudo residuals** $\tilde{r}^{[m]}$.

COMPONENTWISE BOOSTING ALGORITHM

We restrict the base learners to additive model components, i.e.,

Different from GB, multiple base learners $b_j \in \mathcal{B}_j$, $j = 1, \ldots, J$, are fitted and only best-fitting one is selected and updated.

Algorithm Componentwise Gradient Boosting

1: Initialize $f^{[0]}(\mathbf{x}) = \arg\min_{\theta_0 \in \mathbb{R}} \sum_{i=1}^{n} L(y^{(i)}, \theta_0)$

2: **for** $m = 1 \to M$ **do**

3: For all i: $\tilde{r}^{[m](i)} = \left[\frac{\partial L(y, f)}{\partial f}\right]_{f=f^{[m-1]}(\mathbf{x}^{(i)}), y=y^{(i)}}$

4: **for** $j = 1 \to J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

6: $\hat{\theta}_j^{[m]} = \arg\min_{\theta \in \Theta_j} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

8: $j^{[m]} = \arg\min_j \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$

9: Update $f^{[m]}(\mathbf{x}) = f^{[m-1]}(\mathbf{x}) + \alpha \hat{b}_j(\mathbf{x}, \hat{\theta}_{j^{[m]}}^{[m]})$

10: **end for**

11: Output $\hat{f}(\mathbf{x}) = \hat{f}^{[M]}(\mathbf{x})$

non-linear (spline) effect

dummy encoded linear model of a cat. feature

tensor product spline for interaction modelling (e.g. spatial effects)

More advanced base learners could also be Markov random fields, random effects, or trees.

(Same as for GB, New inner loop for CWB)

COMPONENTWISE BOOSTING ALGORITHM

Two BLs of the same type can simply be added by adding up their parameter vectors:

$$b_j(\mathbf{x}, \theta^{[1]}) + b_j(\mathbf{x}, \theta^{[2]}) = b_j(\mathbf{x}, \theta^{[1]} + \theta^{[2]}).$$

Algorithm Componentwise Gradient Boosting (inner loop).

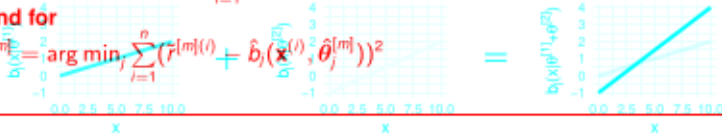
4: **for** $j = 1 \rightarrow J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

6: $\hat{\theta}_j^{[m]} = \arg\min_{\theta \in \Theta_j} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

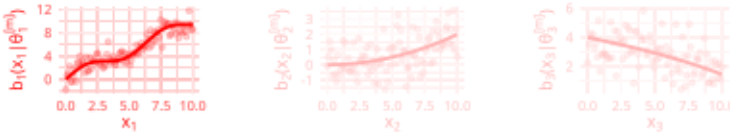
7: **end for**

8: $j^{[m]} = \arg\min_j \sum_{i=1}^{n} (\tilde{r}^{[m](i)} + \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$



Iteration m, $j = 1$, $\sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_1(x_1^{(i)}, \hat{\theta}_1^{[m]}))^2 = 24.4$:

Thus, if $\{b_j(\mathbf{x}, \theta^{[1]}), b_j(\mathbf{x}, \theta^{[2]})\} \in \mathcal{B}_j$, then $b_j(\mathbf{x}, \theta^{[1]} + \theta^{[2]}) \in \mathcal{B}_j$.



COMPONENTWISE BOOSTING ALGORITHM

Different from GB, multiple base learners $b_j \in \mathcal{B}_j$, $j = 1, \ldots, J$, are fitted and only

Algorithm Componentwise Gradient Boosting (inner loop).

Algorithm Componentwise Gradient Boosting.

1: Initialize $f^{[0]}(\mathbf{x}) = \arg\min_{c \in \mathbb{R}} \sum_{i=1}^{n} L(y^{(i)}, c)$

2: **for** $m = 1 \to M$ **do**

3: For all i: $\tilde{r}^{[m](i)} = -\left[\frac{\partial L(y,f)}{\partial f}\right]_{f=f^{[m-1]}(\mathbf{x}^{(i)}), y=y^{(i)}}$

4: **for** $j = 1 \to J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

6: $\hat{\theta}_j^{[m]} = \arg\min_{\theta \in \Theta_j} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

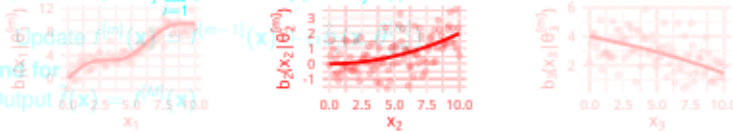
8: $j^{[m]} = \arg\min_j \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$

9: Update $f^{[m]}(\mathbf{x}) = f^{[m-1]}(\mathbf{x}) + \nu \cdot \hat{b}_{j^{[m]}}(\mathbf{x}, \hat{\theta}_{j^{[m]}}^{[m]})$

10: **end for**

11: Output $\hat{f}^{[M]}(\mathbf{x})$

Iteration m, $j = 2$, $\sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_2(x_2^{(i)}, \hat{\theta}_2^{[m]}))^2 = 43.2$:



(Same as for GB. New inner loop for CWB)

COMPONENTWISE BOOSTING ALGORITHM

Algorithm Componentwise Gradient Boosting (inner loop).

4: **for** $j = 1 \rightarrow J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

6: $\hat{\theta}_j^{[m]} = \arg\min_{\theta \in \Theta_j} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

8: $j^{[m]} = \arg\min_j \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$

Iteration m, $j = 3$, $\sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_3(x_3^{(i)}, \hat{\theta}_3^{[m]}))^2 = 35.2$:

COMPONENTWISE BOOSTING ALGORITHM

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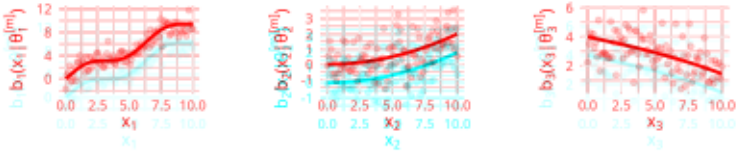
7: **end for**

8: $j^{[m]} = \arg\min_j \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$

Iteration m: $\Rightarrow j^{[m]} = 2$, $\sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_2(x_2^{(i)}, \hat{\theta}_2^{[m]}))^2 = 43.2$:



FEATURE SELECTION IN CWB ALGORITHM

In CWB, we often define BLs on a single feature

Algorithm Componentwise Gradient Boosting (inner loop).

4: **for** $j = 1 \to J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

6: $\hat{\theta}_j^{[m]} = \arg\min_{\theta \in \Theta_j} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(x^{(i)}, \theta))^2$

7: **end for**

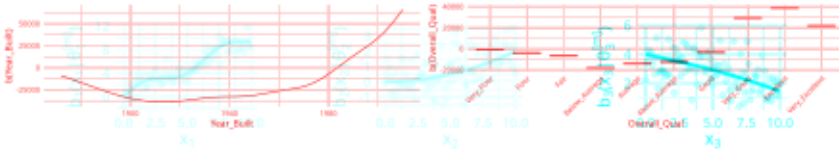
8: $j^{[m]} = \arg\min_{j \in \{1,\dots,J\}} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(x^{(i)}, \hat{\theta}_j^{[m]}))^2$

$b_i(x_i, \theta)$ for $j = 1, 2, \dots, p$.

Allows natural form of feature selection:

- When we select the best BL in one iter of CWB, we thereby also only select one (associated) feature
- Note that a feature (or rather a BL associated with it) can be selected in multiple iters, so $\leq M$ features are selected

Iteration m, $j = 3$, $\sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_3(x_3^{(i)}, \hat{\theta}_3^{[m]}))^2 = 35.2$:



COMPONENTWISE BOOSTING ALGORITHM

Algorithm Componentwise Gradient Boosting (inner loop).

for $j = 1 \to J$ **do**

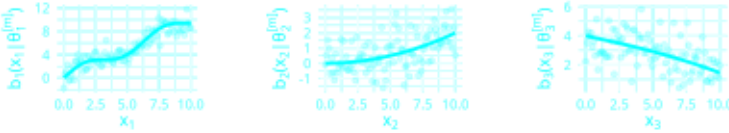
  Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

  $\hat{\theta}_j^{[m]} = \arg\min_{\theta \in \Theta_j} \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

end for

$j^{[m]} = \arg\min_j \sum_{i=1}^{n} (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$

Iteration m: $\Rightarrow j^{[m]} = 1$



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$$b_j(x_j, \theta) \quad \text{for } j = 1, 2, \ldots, p.$$

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